

PRATHAMESH AKOLE

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Available from: Spring 2025

EDUCATION

Northeastern University - Khoury College of Computer Sciences
Master of Science in Computer Science

Sep 2023 – Present
GPA: 3.91 Boston, MA

Savitribai Phule Pune University
Bachelor of Technology in Information Technology

Aug 2019 – Jun 2023
Pune, MH, India

TECHNICAL SKILLS

Languages: Java, C, C++, Python, HTML, SCSS, JavaScript

Frameworks: React.js, Node.js, Express.js, Spring Boot, JUnit, Hibernate

Databases: SQL, MySQL, MongoDB, PostgreSQL, DynamoDB, Redis, GraphQL, RabbitMQ

Cloud DevOps: Amazon Web Services (AWS,EC2,RDS), Docker, Kubernetes, CI/CD (Jenkins), Terraform,Apache Kafka

Others: RESTful APIs, Microservices, MVC, Grafana, Datadog, Git, Agile Methodologies, Test-Driven Development (TDD), Machine Learning(Computer Vision), Linux, Bash, QT

WORK EXPERIENCE

IdeaS Revenue Solutions

Jan 2023 – Jun 2023

Associate Software Developer Intern

Pune, MH

- Developed and executed new features using **Java, Spring** and **RESTful APIs**, resulting in a 15% decrease in post-release bugs reported by the clients
- Engineered forecasting system utilizing **Spring Data JPA, AWS RDS**, and native SQL queries, deploying on AWS EC2 for scalable business analytics
- Optimized complex **SQL** DB queries, reducing execution time by 50% and enhancing data retrieval efficiency
- Spearheaded test-driven development using **JUnit, Mockito, and TestRestTemplate**, delivering unit, integration, and end-to-end tests that increased code coverage to 100%.

PowerMech Engineering Systems

Software Developer Engineer Intern

Pune, MH

- Led a 5-member team to develop a **full-stack** rental platform using **ReactJS, NodeJS** with ML features, achieving 100% WCAG accessibility compliance.
- Implemented **CI/CD** testing pipelines that reduced deployment time by 60% & increased test coverage to 95%.
- Developed ML-based prediction tools in **Python** that achieved 85% accuracy in rental price forecasting with self-created database in **MongoDB**.
- Developed ML-based prediction tools using **Python** with automated accuracy testing and explored **LLM** API integration for enhanced user interactions

PROJECTS

Pinpoint - A Geoguesser Social Media

- * Architected and developed a full-stack application using **React.js,Node.js, ExpressJs with TypeScript**, implementing automated testing pipelines.
- * Deployed and maintained scalable infrastructure using **Github Pages and Onrender**.
- * Implemented secure **user authentication** and content moderation systems, focusing on accessibility and **UI/UX**

Distributed Key-Value Store with Fault Tolerance

- * Architected a distributed key-value store using **Java RMI**, and PAXOS consensus protocol achieving 99.9% data consistency across 5+ replicated nodes while maintaining fault tolerance with < 100ms latency
- * Containerized the app using **Docker** and automated deployment through **Bash** scripting, streamlining the workflow.
- * Engineered a fault-tolerant multi-threaded system achieving 100% availability with 3-second recovery during simultaneous node failures
- * Implemented **multithreading** in **Java** to handle concurrent client requests, improving system throughput by 50%

TECHNICAL PUBLICATIONS

- * Real-Time CCTV Violence Detection System Using Deep Learning, April 2023, ISBN: 978-1-6654-7524-2.
- * Real-Time CNN-Based Face Mask Detection System, December 2022, ISBN: 9781032374598.